

Operator Method

$$\textcircled{1} \quad \frac{1}{(D-2)} r(x) = e^{2x} \int e^{-2x} r(x) dx$$

$$\textcircled{2} \quad \frac{1}{f(D)} e^{ax} = \frac{1}{f(a)} e^{ax} \quad \text{if } f(a) \neq 0$$

$$\textcircled{3} \quad \textcircled{a} \quad \frac{1}{(D-2)^n} e^{2x} = \frac{x^n}{n!} e^{2x}$$

$$\textcircled{b} \quad \frac{1}{(D-2)^k q(D)} = \frac{1}{q(2)} \cdot \frac{x^k}{k!} e^{2x} \quad \text{if } q(2) \neq 0$$

$$\textcircled{4} \quad \textcircled{a} \quad \frac{1}{\varphi(D^2)} \sin(ax) = \frac{1}{\varphi(-a^2)} \sin(ax) \quad \text{if } \boxed{\varphi(-a^2) \neq 0}$$

$$\textcircled{b} \quad \frac{1}{\varphi(D^2)} \cos(ax) = \frac{1}{\varphi(-a^2)} \cos(ax) \quad \text{if } \varphi(-a^2) \neq 0$$

① Solve

$$(D^2 + a^2) y = \sin(ax)$$

P.I

$$\frac{1}{D^2 + a^2} \sin(ax)$$

Euler's formula

$$e^{ian} = \cos(ax) + i \sin(ax)$$

P.I

$$\boxed{\frac{1}{D^2 + a^2} e^{ian}}$$

Real part

Imaginary part

$$\frac{1}{D^2 + a^2} \cos(ax)$$

$$\frac{1}{D^2 + a^2} \sin(ax)$$

$$\frac{1}{D^2 + a^2} e^{ian} = \frac{1}{(D - ia)(D + ia)} e^{ian} = \frac{1}{2ai} \cdot \frac{x}{1!} e^{ian}$$

$$\left[\frac{1}{(D-2)^k q(D)} e^{ax} = \frac{1}{q(a)} \cdot \frac{x^k}{k!} e^{ax} \right] \quad q(a) \neq 0$$

$$\cdot \frac{-i}{2a} (x \cos ax + i x \sin ax) = \frac{x \sin(ax)}{2a} - \frac{i x \cos ax}{2a}$$

$$\star \frac{1}{b^2 + a^2} \sin ax = \frac{-x \cos(ax)}{2a}$$

• Exponential shift Method

$$\cdot \frac{1}{f(D)} e^{\alpha x} x(x) = e^{\alpha x} \boxed{\frac{1}{f(D+\alpha)} x(x)}$$

$$\star D(e^{\alpha x} x(x)) = e^{\alpha x} D(x(x)) + \alpha e^{\alpha x} x(x) = e^{\alpha x} (D+\alpha)(x(x))$$

$$\star D^2(e^{\alpha x} x(x)) = e^{\alpha x} (D+\alpha)^2 x(x)$$

$$(D-\alpha)y=0$$

• Solve $(D-1)y = e^x \sec^2 x \tan x$

P.I. $\frac{1}{D-1} e^x \sec^2 x \tan x = e^x \frac{1}{(D+1)-1} \sec^2 x \tan x$

$$= e^x \frac{1}{D} \sec^2 x \tan x = e^x \int \sec^2 x \tan x dx$$

$$\tan x = z$$

$$\sec^2 x dx = dz$$

$$= \frac{e^x (\tan x)^2}{2}$$

$$\star \frac{1}{f(D)} x^m \sin(ax)$$

OR

$$\frac{1}{f(D)} x^m \cos(ax)$$

$$\cdot \frac{1}{f(D)} x^m e^{i a x}$$

$$e^{i a x} = \cos(ax) + i \sin(ax)$$

Real part

$$\frac{1}{f(D)} x^m \cos(ax)$$

Imaginary part

$$\frac{1}{f(D)} x^m \sin(ax)$$

series method

$$\frac{1}{f(D)} x^m \cos(ax) \quad \overline{f(D)} \quad \text{series method}$$

$$\frac{1}{f(D)} e^{iax} x^m = e^{iax} \frac{1}{f(D+ia)} x^m = e^{iax} (1 + \phi(D+ia))^{-1} x^m$$

Question:- Solve $(D^2+9)y = x \sin x$.

P.I. $\frac{1}{D^2+9} (x \sin x)$

Imaginary part of $\frac{1}{D^2+9} x e^{ix}$

$$\begin{aligned} \frac{1}{D^2+9} e^{ix} x &= e^{ix} \frac{1}{(D+i)^2+9} x = e^{ix} \frac{1}{D^2-1+2Di+9} x \\ &= e^{ix} \frac{1}{8+D^2+2Di} x = \frac{e^{ix}}{8} \left(1 + \left(\frac{D^2+2Di}{8}\right)\right)^{-1} x \\ &= \frac{e^{ix}}{8} \left(1 - \frac{D^2}{8} - \frac{2Di}{8}\right) x = \frac{e^{ix}}{8} \left(x - \frac{i}{4}\right) \\ &= \frac{(\cos x + i \sin x)}{8} \left(x - \frac{i}{4}\right) \end{aligned}$$

Imaginary part is $\rightarrow \frac{x \sin x}{8} - \frac{\cos x}{32}$

Euler Cauchy Equation:-

$$x^2 y'' + ax y' + by = 0$$

a, b are some constants.

$$y = x^m$$

$$y' = m x^{m-1}$$

$$y'' = m(m-1) x^{m-2}$$

Using eqn^s we get

$$m(m-1)x^m + amx^m + bx^m = 0$$

Very nice

$$m(m-1)x^m + amx^m + bx^m = 0$$

$$x^m (m(m-1) + am + b) = 0$$

$$\Rightarrow m(m-1) + am + b = 0$$

$$\Rightarrow m^2 + (a-1)m + b = 0$$

$$m_1 \neq m_2$$

Two distinct
real roots
 m_1, m_2

$$y = C_1 x^{m_1} + C_2 x^{m_2}$$

Single
real root
 m_1, m_1

$$x^{m_1}$$

Reduction of order

two complex
roots
 $\alpha \pm i\beta$

$$C_1 x^{\alpha+i\beta} + C_2 x^{\alpha-i\beta}$$

Case (iii)

Roots are $\alpha \pm i\beta$

$$y = C_1 x^{\alpha+i\beta} + C_2 x^{\alpha-i\beta} = x^{\alpha} (C_1 x^{i\beta} + C_2 x^{-i\beta})$$

$$y = x^{\alpha} (C_1 e^{\log(x^{i\beta})} + C_2 e^{\log(x^{-i\beta})})$$

$$y = x^{\alpha} (C_1 e^{i\beta \log x} + C_2 e^{-i\beta \log x})$$

$$y = x^{\alpha} (C_1 (\cos(\beta \log x) + i \sin(\beta \log x)) + C_2 (\cos(\beta \log x) - i \sin(\beta \log x)))$$

$$y = x^{\alpha} (\underbrace{C_1 + C_2}_{K_1} (\cos(\beta \log x)) + i \underbrace{(C_1 - C_2)}_{K_2} \sin(\beta \log x))$$

$$y = x^{\alpha} (K_1 \cos(\beta \log x) + K_2 \sin(\beta \log x))$$

If roots are $\alpha \pm i\beta$

$$m^2 + (a-1)m + b = 0$$

$$D = (a-1)^2 - 4b = 0$$

$$\Rightarrow b = \frac{(a-1)^2}{4}$$

$$D = (a-1)^2 - 4b = 0 \Rightarrow b = \frac{(a-1)^2}{4}$$

$$0 = m^2 + (a-1)m + \frac{(a-1)^2}{4} = \left(m + \frac{a-1}{2}\right)^2$$

$$m = \frac{1-a}{2}$$

$$x^2 y'' + a x y' + b y = 0, \quad y_1 = x^m, \quad \text{where } m = \frac{1-a}{2}$$

$$y_2 = u(x) x^m$$

$$y_2' = u(x) m x^{m-1} + u'(x) x^m$$

$$y_2'' = 2 u'(x) m x^{m-1} + u(x) m(m-1) x^{m-2} + u''(x) x^m$$

$$2 u'(x) m x^{m+1} + \underline{u(x) m(m-1) x^m} + u''(x) x^{m+2} + \underline{a m u(x) x^m} + a u'(x) x^{m+1} + \underline{b u(x) x^m} = 0$$

$$u(x) x^m \underbrace{\left(m(m-1) + a m + b \right)}_{\substack{|| \\ 0}} + u'(x) x^{m+1} \underbrace{\left(2m + a \right)}_{\substack{\downarrow \\ m = \frac{1-a}{2} \\ \hline 2m+a=1}} + u''(x) x^{m+2} = 0$$

$$\Rightarrow x^{m+1} (u'(x) + u''(x) x) = 0$$

$$\Rightarrow u'(x) + u''(x) x = 0$$

put $u'(x) = \phi(x)$

$$\Rightarrow \phi(x) + \phi'(x) x = 0$$

$$\Rightarrow \phi + \frac{d\phi}{dx} x = 0 \Rightarrow \int \frac{d\phi}{\phi} = \int -\frac{dx}{x}$$

$$\Rightarrow \log \phi = -\log x \Rightarrow \phi = \frac{1}{x}$$

$$u'(x) = \frac{1}{x} \Rightarrow u(x) = \int \frac{1}{x} dx = \log x$$

$$u'(u) = \frac{1}{u} \Rightarrow u(u) = \int \frac{1}{u} du = \log u$$

$$y_2 = \log u \cdot u^m, \quad m = \frac{1-a}{2}$$

• u^m & $\log u \cdot u^m \rightarrow$ L.I. solution in case of repeated roots.

• $u^2 y'' + a u y' + b y = 0 \rightarrow$ Euler Cauchy Eqn
Aux. Eqn $\rightarrow m(m-1) + am + b = 0$ OR $m^2 + (a-1)m + b = 0$
Find roots of Aux Eqn

Real $m_1, m_2, m_1 \neq m_2$
 $C_1 u^{m_1} + C_2 u^{m_2}$

Real m_1, m_2
 $C_1 u^{m_1} + C_2 \log u \cdot u^{m_2}$

$\alpha \pm i\beta$
 $u^\alpha (C_1 \cos(\beta \log u) + C_2 \sin(\beta \log u))$

① $u^2 y'' - 3u y' + 4y = 0$

Aux Eqn $\rightarrow m(m-1) - 3m + 4 = 0 \Rightarrow m^2 - 4m + 4 = 0$
 $\Rightarrow m = 2, 2$

Solution $y = C_1 u^2 + C_2 \log u \cdot u^2$

② $u y'' + 2y' = 0 \Rightarrow u^2 y'' + 2u y' + 0y = 0$

Aux Eqn $m(m-1) + 2m + 0 = 0$
 $m^2 + m = 0 \Rightarrow m = 0, -1$

Solution $y = C_1 u^0 + C_2 u^{-1} = C_1 + \frac{C_2}{u}$

③ $u^2 y'' + 7u y' + 13y = 0$

Aux Eqn

$$m(m-1) + 7m + 13 = 0$$

$$m^2 - 6m + 13 = 0$$

$$m = -3 \pm 2i$$

$$y = x^{-3} \left(c_1 \cos(2 \log x) + c_2 \sin(2 \log x) \right)$$